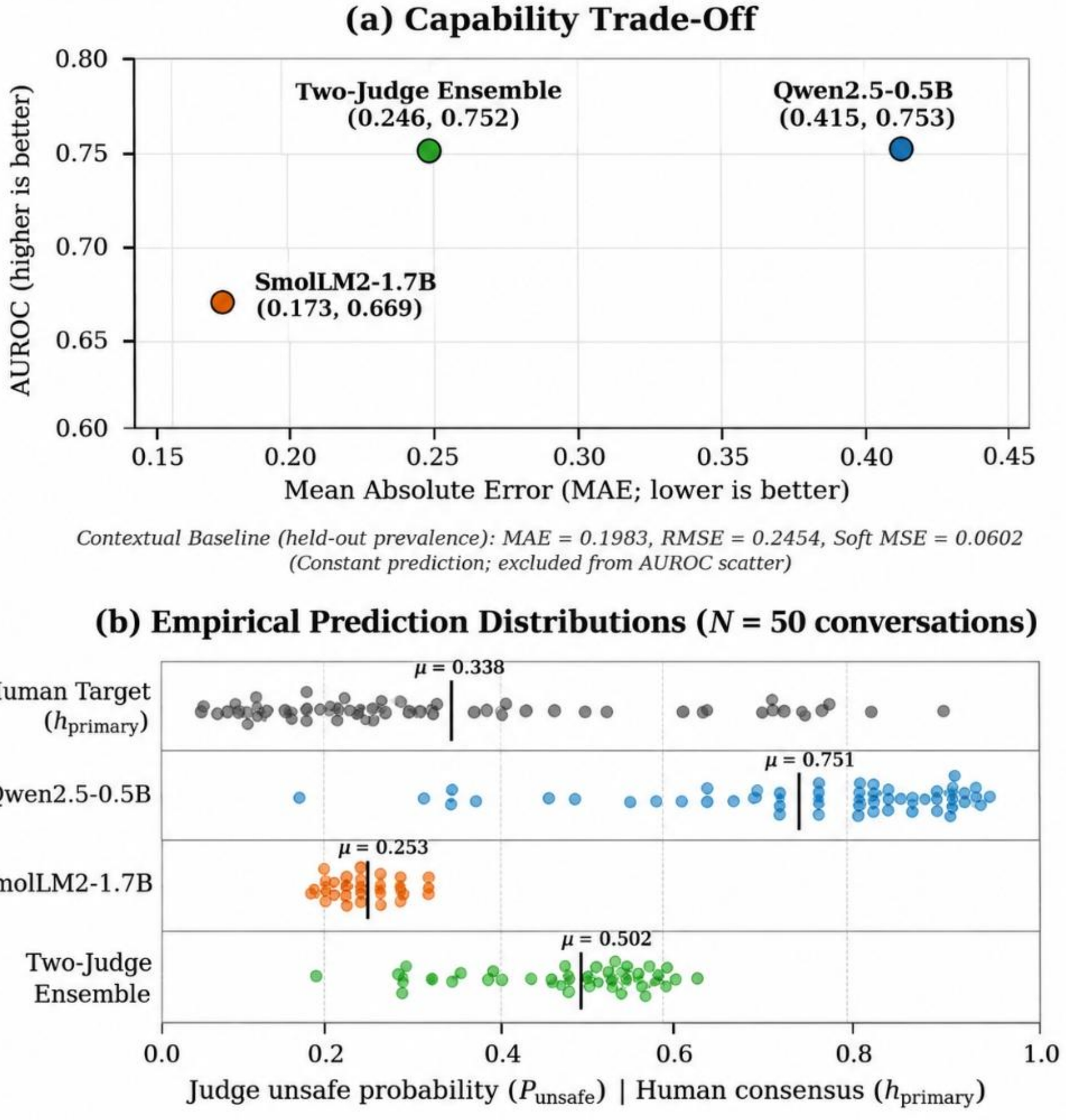

(a) Capability Trade-Off
AUROC (higher is better)
0.80
0.75
0.70
0.65
0.60
Two-Judge Ensemble
(0.246, 0.752)
Qwen2.5-0.5B
(0.415, 0.753)
SmolLM2-1.7B
(0.173, 0.669)
0.15
0.20
0.25
0.30
0.35
0.40
0.45
Mean Absolute Error (MAE; lower is better)
Contextual Baseline (held-out prevalence): MAE = 0.1983, RMSE = 0.2454, Soft MSE = 0.0602
(Constant prediction; excluded from AUROC scatter)
(b) Empirical Prediction Distributions (N = 50 conversations)
Human Target
($h_{primary}$)
Qwen2.5-0.5B
SmolLM2-1.7B
Two-Judge
Ensemble
μ = 0.338
μ = 0.751
μ = 0.253
μ = 0.502
0.0
0.2
0.4
0.6
0.8
1.0
Judge unsafe probability (P_{unsafe}) | Human consensus ($h_{primary}$)